Joseph Haas

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Senior Software Engineer | Technical Product Manager

Backend & API Development ▪ Payments & Benefits Systems ▪ Product Strategy & Delivery

Senior software engineer with 5 years of experience building production systems for payments and ledger platforms that fund, track, and reconcile $1B annually in benefits for Fortune 1000 employer partners and 3.37M learners, plus 8 years of experience as a product manager shipping enterprise software. Specializing in financial systems, combining API design, distributed architecture, and cross-functional delivery to bridge engineering, product, and operations.

Areas of Expertise

- *Backend & API Design*
- *Event-Driven Architecture*
- *System Architecture & ADRs*
- *Performance & Load Testing*
- *Product Strategy & Roadmaps*
- *Payments & Ledger Systems*
- *Data Modeling & Database Design*
- *Observability & Alerting*
- *Security Remediation*
- *Agile, BDD & SAFe Delivery*
- *Distributed Systems*
- *Data Migrations & Backfills*
- *Incident Response & Forensics*
- *B2B & Benefits Integration*
- *Technical Documentation & Runbooks*

Technical Skills: Node.js, TypeScript, Python, SQL, PostgreSQL, Snowflake, REST API development, GraphQL (AWS AppSync), event-driven architecture, AWS (Lambda, AppSync, RDS), containerization, Git, GitHub Actions (CI/CD), Mocha/Chai/Sinon, integration testing, BDD, Snyk, Salesforce and NetSuite integrations, Jira, Confluence

Professional Experience

GUILD — Denver, CO 2020–Present

Education and career-mobility benefits platform serving Fortune 1000 employers. The payments engineering team owns the systems that fund, track, and reconcile roughly $1B annually in benefits for 3.37M learners.

Senior Software Engineer *2024–Present*

- Led the migration of the ledger system—serving employer partners and 3.37M learners—to a new eligibility platform. Defined the API contract and authored the technical specification, then rewrote tax-treatment and spending-control logic to align business rules across systems. Built a production side-by-side parity framework with request/response validation and state reconciliation checks, and executed a phased cutover with zero rollbacks or incidents.
- Built a bulk balance calculation endpoint supporting up to 500 accounts per call, replacing thousands of individual requests. Refactored it to fetch benefit configuration once per batch, cutting mean latency by 55–74% and increasing throughput by 2–4× for batches of 50–500 accounts in production. Automated the load tests used to measure these improvements.
- Drove the research into and adoption of event-driven integration for secure, automated partner data ingestion. Defined the event schema and the integration rollout plan.
- Built core features for partner calendar management system, including cross-system data write, a business rules engine, and integration test infrastructure.
- Strengthened service availability by building metric-based alerting and runbooks to detect spend-check failures, automating deployment validation with scripted smoke tests, and remediating critical vulnerabilities.
- Supported incident investigations and data forensics across the team's service and Snowflake, documenting findings and procedures in reusable runbooks.

Software Engineer II *2022–2024*

- Architected and shipped createSpend, the core spend-writing API behind about $1B in annual spending across four benefits. Designed a single-write-path API with unified request and response models, replaced fragmented legacy patterns with canonical data models, added idempotency safeguards to prevent duplicate writes, and built partner-aware funding logic with deterministic funding determination.
- Designed the validation and benefit utilization correction service from inception, building endpoints that carry internal identifiers, external partner identifiers, and eligibility context into every spend decision.
- Built features supporting an external reporting system end-to-end, including the backend API and frontend, enabling self-service reporting for B2B partners.
- Helped build a new core platform foundation, including authorization scopes, read replicas, a bulk update endpoint, and an event-driven consumer. Implemented dual writes for adjustment and data-merge events to support ledger-accurate spend tracking; authored the backfill and migration communication strategy.

Software Engineer I *2021–2022*

- Delivered the Salesforce reconciliation integration, enabling accurate end-of-year benefit accounting and preventing employer overpayments and underpayments across fiscal years.
- Extended the team's GraphQL APIs with filtering, pagination, bulk writes, and tax-period endpoints, and integrated them into the company's Federated GraphQL gateway.

- Built an internal admin tool for bulk spend corrections so Operations could correct data without engineering involvement.
- Onboarded nine employer partners through benefit configuration and SQL data migrations, wrote runbooks that shortened subsequent launches, and migrated security scanning from ShiftLeft to Snyk across five repositories.

Product Manager, Payment Products *2020–2021*

- Owned balance-tracking and external benefits administration products for cross-functional payments scrum teams, supporting Fortune 1000 employer partners and the systems that determine how each learner's benefit dollars are funded and reconciled.
- Wrote detailed user stories and frontend mockups, defined MVPs, and prioritized the backlog for engineering teams delivering payments features on a two-week cadence.
- Coordinated multi-team projects with fellow product managers and engineering leaders, aligning requirements, deployment sequencing, and cross-squad dependencies for releases spanning several services.

MJ FREEWAY — Denver, CO 2018–2019

Software Product Manager, Core Services

- Served as product owner for the Core Services suite (Cultivation, Processing, and Point of Sale). Defined the end-to-end enterprise development strategy using the SAFe framework and maintained a release cadence of two weeks.
- Wrote requirements, user stories, wireframes, and UI mockups; collaborated with enterprise clients to validate roadmap milestones and reduce refactoring.
- Trained the product organization on SDLC practices, from writing bug reports and user stories to identifying core customer needs; led the cross-functional evaluation of new global markets.

CHARTER COMMUNICATIONS — Denver, CO 2014–2018

Manager, Video Products

- Defined the roadmap and owned the backlog for three TiVo set-top video guide products supporting 20% of enterprise video customers. Managed features across set-top boxes, Roku, Xbox, Samsung, web, and mobile platforms.
- Served as feature owner for accessibility compliance, partnering with UX to deliver a text-to-speech guide usable without vision, a preinstalled Samsung tablet app, and a web portal with cloud DVR.
- Won the Charter Champion Award (2015) for rebuilding internal and customer-facing support tools with the Care team, reducing call handling time and adding self-service options.

iBAHN — Denver, CO 2012–2014

Associate Product Manager, IPTV Platform / Hospitality Public PC Solutions

- Managed the ETVI in-room video platform for hotels, including custom set-top hardware, live TV and on-demand interfaces, and room-control integrations. Led marketing, business development, and global product development for a public-PC product for hotel lobbies and business centers.
- Gathered UI, custom-skin, and feature requirements for large clients, including Marriott, Hilton, and Four Seasons; translated customer issues into feature requirements and improvements to support and error handling.
- Analyzed market trends and competitor data to identify product gaps; directed an internship team conducting competitive research for the ETVI product.

Education & Certifications

Back-End Engineering Program | Turing School of Software and Design, Remote

Bachelor of Science in International Business, Minor in Marketing | University of Denver, Denver, CO

Certifications: Certified Scrum Product Owner (Scrum Alliance) ▪ Pragmatic Marketing Certification (Pragmatic Institute)